

1. Administrative & Study Identification

Full Study Title: A Multicentre, Open-label, Randomised Phase IV Study to Investigate Acalabrutinib Monotherapy Compared to Investigator's Choice of Treatment in Adults (> 18 Years) With Chronic Lymphocytic Leukaemia and Moderate to Severe Cardiac Impairment **Short Title/Acronym:** ELEVATE-CV **Protocol Number:** D8223C00016 **NCT Number:** NCT06651970 **Posting ID:** 388178163294466888 **Trial Status:** RECRUITING **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Acalabrutinib (Trade Name: Calquence) **Phase of Development:** Phase IV **Medical Monitor:** AstraZeneca Clinical Study Information Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective (Safety): To evaluate the incidence of cardiovascular (CV) adverse events leading to drug discontinuation, the duration on treatment prior to drug discontinuation due to CV adverse events, the incidence of Grade 4 and 5 cardiovascular events of interest, and the incidence of life-threatening and fatal cardiac events (including Grade ≥ 3 AEs) after acalabrutinib treatment compared to investigator's choice of treatment. **Secondary Objectives (Efficacy):** To evaluate overall survival (OS), overall response rate (ORR), and duration of response (DoR) after acalabrutinib treatment compared to investigator's choice of treatment.

2.2 Background & Rationale To evaluate the safety, tolerability, and efficacy of acalabrutinib monotherapy (100 mg twice daily) compared to the investigator's choice of treatment in patients with Chronic Lymphocytic Leukaemia (CLL) (treatment-naïve or relapsed/refractory) who concurrently suffer from moderate to severe cardiac impairment (Left Ventricular Ejection Fraction [LVEF] $\leq 50\%$).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Investigator's Choice of Treatment) **Blinding:** Open-label **Randomization:** Randomized (Stratified by LVEF: $> 40\%$ vs $\leq 40\%$)

3.2 Planned Enrollment & Duration Target N: 60 **Number of Sites:** Approximately 20 centres globally **Estimated Study Start:** 04-Feb-2025 **Estimated Primary Completion:** 16-Aug-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults (> 18 years) diagnosed with Chronic Lymphocytic Leukaemia (CLL) (TN or R/R). 2. Moderate to severe cardiac impairment, defined as LVEF $< 50\%$ (Severe: LVEF $\leq 40\%$; Moderate: LVEF $> 40\%$ to $< 50\%$).

4.2 Exclusion Criteria 1. Uncontrolled cardiac tachyarrhythmias requiring new/additional therapy within the last month, or QTcF > 470 ms or < 330 ms. 2. Use of strong CYP3A inhibitors within 1 week or strong CYP3A inducers within 3 weeks of the first dose of study treatment. 3. Concurrent participation in another therapeutic clinical trial, or currently pregnant/breastfeeding.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Acalabrutinib 100 mg orally twice daily (bd), OR Investigator's choice of treatment (chlorambucil, venetoclax, ibrutinib, zanabrutinib, rituximab, obinutuzumab, etc.). **Route of Administration:** Oral (for acalabrutinib) or standard route for the chosen comparator. **Duration of Treatment:** Treatment is administered until unacceptable toxicity or disease progression.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for comorbidities not explicitly restricted by the protocol. **Prohibited:** Strong CYP3A inhibitors or inducers, and live virus vaccines within 28 days of the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening	Up to Day 0 Informed Consent, Medical History, LVEF baseline testing
2-9	Interim	Cycles 1 to 8 (every 4 weeks) Haematology labs, physical exams (Day 1 of each 28-day cycle)
10-17	Long-term Follow-up	Every 4 cycles (every 16 weeks) Routine labs, efficacy assessment, cardiology assessments (including cardiology consults)
18	End of Study	Follow-up Phase Safety Follow-Up (SFU) within 45 days of last dose: Cardiology consult, ECG, ECHO, Holter, Cardiac MRI

(Note: Study termination occurs 4 years from the last subject randomized.)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints (Safety) - Safety Endpoint 1: Incidence of CV adverse events leading to drug discontinuation (frequency and

time to discontinuation of any study treatment due to worsening in cardiovascular function or cardiovascular AEs). - **Safety Endpoint 2:** Duration on treatment prior to drug discontinuation due to CV adverse events and the incidence of Grade 4 and 5 cardiovascular events of interest. - **Safety Endpoint 3:** Incidence of life-threatening and fatal cardiac events of interest, including the incidence and relationship to study treatment of Grade ≥ 3 AEs.

7.2 Secondary Endpoints (Efficacy) - **Efficacy Endpoint 1:** Overall survival (OS) after acalabrutinib treatment compared to investigator's choice of treatment. - **Efficacy Endpoint 2:** Overall response rate (ORR) after acalabrutinib treatment compared to investigator's choice of treatment. - **Efficacy Endpoint 3:** Duration of response (DoR) after acalabrutinib treatment compared to investigator's choice of treatment.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine safety assessments at every visit. Extensive cardiology assessments (ECHO, ECG, 24-hour Holter, Cardiac MRI). **Serious Adverse Events (SAEs):** Specific monitoring for Grade 4/5 cardiovascular events, fatal cardiac events, and treatment discontinuation due to cardiovascular function worsening. **Data Monitoring Committee (DMC):** An Independent Data Monitoring Committee (IDMC) is responsible for reviewing safety and making recommendations for study continuation.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy endpoints (OS, ORR, DoR), and Safety Set for the evaluation of primary safety endpoints and adverse events. **Sample Size Justification:** A target $N=60$ is considered appropriate for a Phase IV evaluation primarily investigating the safety, tolerability, and comparative survival in a highly specific subgroup (CLL with established heart failure). **Handling of Missing Data:** Standard time-to-event analysis models (e.g., Kaplan-Meier estimation) handling censoring for missing or incomplete follow-up data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Conducted globally in centers establishing close collaboration between Haematology and Cardiology divisions, preferably utilizing an on-site cardio-oncologist. **Audit Trail:**

eCRF locking and compliance tracking will be maintained following standard industry protocols for data integrity.

11. Study Identifiers & Governance

Primary ID: D8223C00016 **Registry Number:** NCT06651970 **Posting ID:** 388178163294466888 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ELEVATE-CV **Official Regulatory Title:** A Multicentre, Open-label, Randomised Phase IV Study to Investigate Acalabrutinib Monotherapy Compared to Investigator's Choice of Treatment in Adults (> 18 Years) With Chronic Lymphocytic Leukaemia and Moderate to Severe Cardiac Impairment

12. Clinical Framework (Study Specific)

Primary Conditions: Chronic Lymphocytic Leukaemia (CLL) and Heart Failure **Diagnosis Threshold:** Moderate to severe cardiac impairment (LVEF $\leq 50\%$) **Medical Methodology:** BTK Inhibitor Therapy vs. Standard Care **Optimization Target:** Cardiovascular event characterization/reduction and overall survival

13. Study Procedures & Methodology

Management Pathway: Interdisciplinary collaboration between Haematology and Cardiology **Education Protocol (HCP):** Cardio-oncologist integration and clinical guidelines monitoring **Education Protocol (Patient):** Safety and reporting protocols for cardiovascular events **Quality Audit Mechanism:** Ongoing safety reviews by an Independent Data Monitoring Committee (IDMC)

14. Observation & Follow-Up Schedule

Total Duration: 4 years from the last subject randomized **Onsite Visit Cadence:** Every 4 weeks (Cycles 1-8), then every 16 weeks **Remote Monitoring Frequency:** As required per standard AE reporting **Checklist Review Interval:** Comprehensive cardiology assessments at the end of Cycle 1, Cycle 3, and then every 16 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	____	[DD-MMM-YYYY]				